# HR Analytics SQL Project Report

**Tools Used:**

- MySQL Workbench 8.0
- Microsoft Excel
- GitHub

**Database:** HR Analytics

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# Project Overview

## Objective

The objective of this project is to analyze employee data using SQL to gain insights into workforce demographics, compensation, employee performance, and attrition. The project demonstrates SQL skills including joins, aggregate functions, window functions, subqueries, and Common Table Expressions (CTEs).

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# Dataset Information

**Dataset:** IBM HR Analytics Employee Attrition

**Source:** Kaggle

**Total Records:** 1,470 Employees

**Tables Created:**

- Employee
- Department
- Compensation
- Performance

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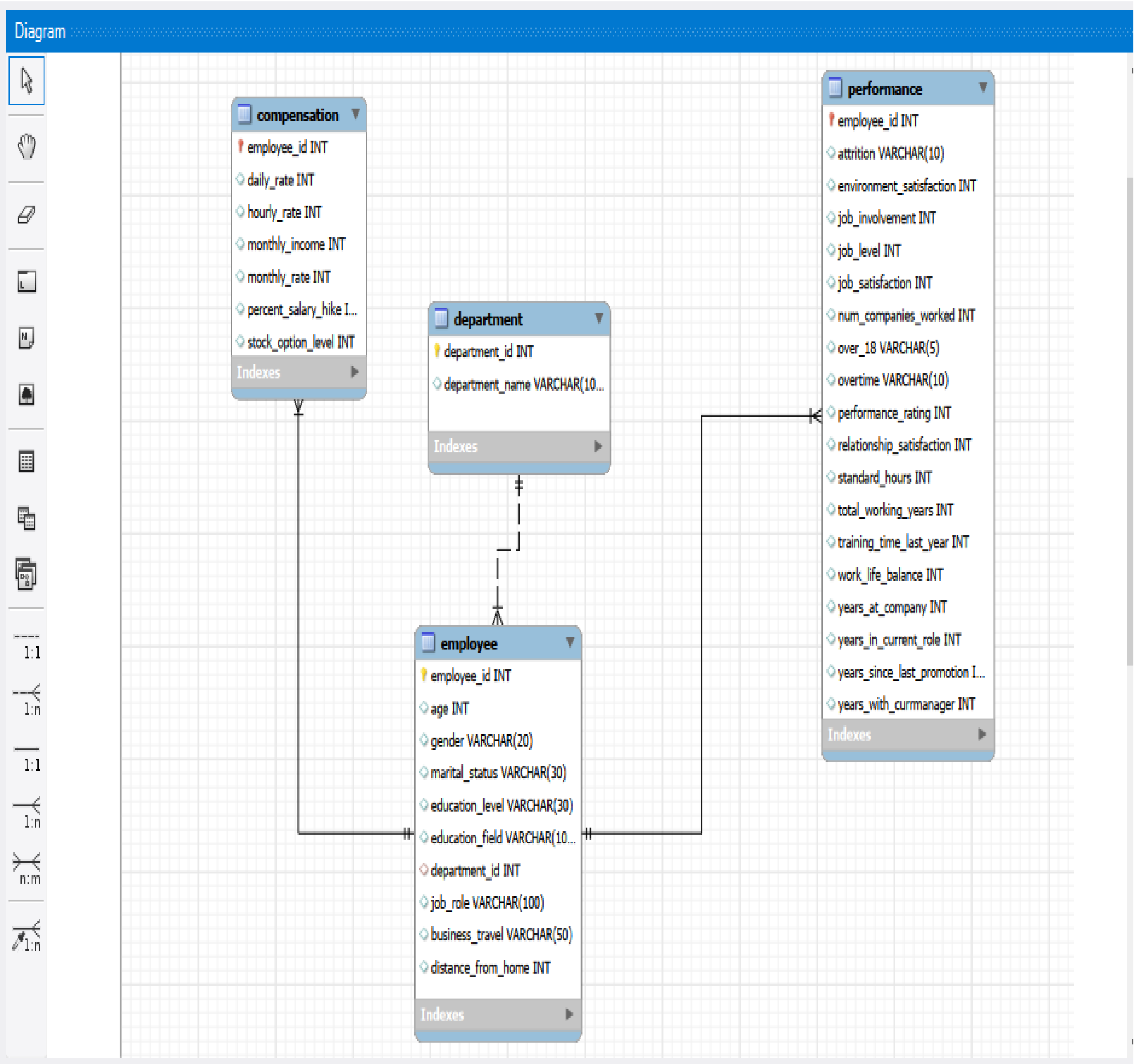
# Database Schema

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# Database Design

### Employees

- Employee ID
- Age
- Gender
- Marital Status
- Education Level
- Education Field
- Department ID

- Job Role
- Business Travel
- Distance From Home

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### Departments

- Department ID
- Department Name

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### Compensation

- Employee ID
- Daily Rate
- Hourly Rate
- Monthly Income
- Monthly Rate
- Salary Hike
- Stock Option Level

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### Performance

- Employee ID
- Attrition
- Performance Rating
- Job Satisfaction
- Work Life Balance
- Total Working Years
- Years At Company
- Overtime ….so no

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# 6. SQL Concepts Implemented

- Aggregate Functions
- GROUP BY
- HAVING
- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- Window Functions
- RANK()
- DENSE_RANK()

- LAG()
- Subqueries

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# Key SQL Queries

Include screenshots of your important queries:

1. Employees by Department
2. Average Salary by Department
3. Top Paid Employees
4. Attrition by Department
5. Rank Employees by Salary
6. Running Total Salary
7. Employees Above Average Salary
8. Departments Above Average Salary
9. LEFT JOIN Example

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# Business Insights

Some example insights:

- Research & Development has the highest number of employees.
- Sales department shows higher employee attrition than Human Resources.
- Employees with higher monthly income generally have lower attrition.
- Overtime is associated with a greater proportion of attrition.
- Certain job roles receive higher average salaries than others.
- Average work-life balance varies across employees and can be monitored alongside attrition.

**Note:** Only include insights that are supported by your actual query results. If your results differ, update these statements accordingly.

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# Project Outcomes

This project demonstrates practical SQL skills including:

- Database Design
- Data Normalization
- SQL Query Writing
- Business Data Analysis
- Reporting

- Data Exploration

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# Conclusion

The HR Analytics SQL Project demonstrates how SQL can be used to analyze employee data and support HR decision-making. By organizing the data into multiple related tables and applying SQL techniques such as joins, subqueries, window functions, meaningful insights about employee demographics, salaries, performance, and attrition can be generated. This project strengthens practical SQL skills and showcases the ability to solve real-world business problems using relational databases.